

WASP-2b

R-1689

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_130822 · created 2026-08-02T12:57:49+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456526.8278 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.047
Transit depth (Rp/Rs)^2	2.39 +/- 1.45 [%]
Orbital Inclination (inc)	86.76 +/- 2.89
Airmass coefficient 1 (a1)	1.29 +/- 0.81
Airmass coefficient 2 (a2)	-0.15 +/- 0.27
Scatter in the residuals of the lightcurve fit is	63.93 %
Best Comparison Star	None
Optimal Aperture	6.69
Optimal Annulus	16.84
Transit Duration (day)	0.082 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.047 vs 0.1326 (deviation 0.48σ)
Duration fit vs published	0.082 d vs 0.0737 d (deviation 0.36σ)
Mid-time O–C	-7.3 min
Depth SNR	3.3
Residual scatter	63.93 %

Light curve

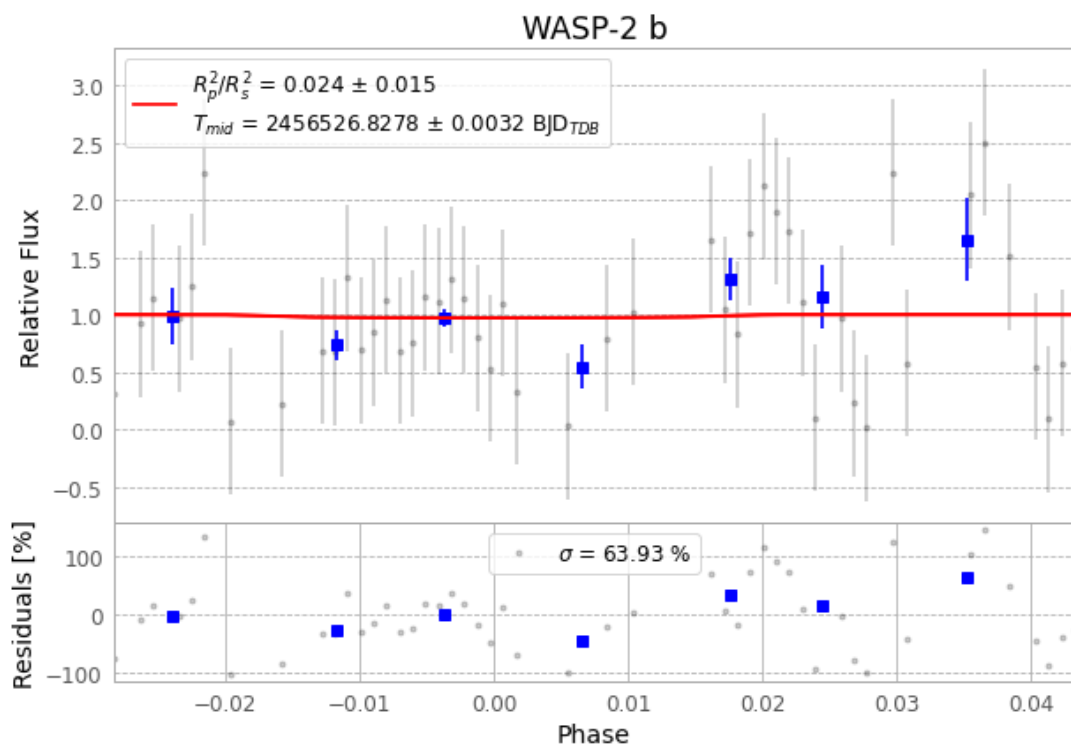


Figure 1 — FinalLightCurve_WASP-2 b_2013-08-21.png (SHA-256 3a638f0b4271260d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [489, 364], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8abd6bf803886a21...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-2130822061214.FITS → WASP-2130822100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-130822.log (b26832944b6d...), FinalLightCurve_WASP-2 b_2013-08-21.png (3a638f0b4271...), FinalLightCurve_WASP-2 b_2013-08-21.csv (5ba09cb1dab3...)

Compute resources

Wall time 335.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 12:57Z.